Churn Prediction and Recommendation System with Uplift Modeling

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Abstract — Customer churn is a critical challenge for businesses, costing billions annually in lost revenue and customer lifetime value. While traditional machine learning models can predict which customers are likely to churn, they fail to identify which retention interventions will be most effective for specific individuals. This project addresses this gap by developing an integrated churn prediction and recommendation system powered by uplift modeling. The system will predict at-risk customers, estimate treatment effects for various interventions, and generate personalized recommendations that maximize retention. By building a production-style architecture with robust data pipelines, validation mechanisms, and monitoring capabilities, this project aims to create a system that reflects real-world deployment standards used in big tech environments.

Team Members & Roles — Team Member: Aryan Atre

Role: Project Lead - Responsible for all aspects of system design, implementation, data pipeline development, model training, validation, and documentation.

I. Problem Statement

Customer churn prediction has become a cornerstone of retention strategy in subscription-based and service-oriented businesses. However, existing approaches suffer from a fundamental limitation: they identify who will churn but provide no guidance on how to prevent it. Traditional models output binary predictions or churn probabilities without accounting for heterogeneous treatment effects—the reality that different customers respond differently to the same intervention.

This project tackles three interconnected problems:

- Predicting customer churn before it occurs, with sufficient lead time to enable intervention
- Identifying which specific retention actions (discounts, feature recommendations, personalized outreach) will be most effective for individual customers
- Optimizing resource allocation by focusing retention efforts on customers most likely to respond positively to intervention

II. Motivation & Significance

The business impact of customer churn is substantial. Studies estimate that acquiring a new customer costs 5-25 times more than retaining an existing one and improving retention rates by just 5% can increase profits by 25-95%. Despite this, most churn models remain prediction-focused rather than action-oriented.

From a technical standpoint, uplift modeling represents a significant advancement over traditional approaches. By estimating causal treatment effects, uplift models can identify customers who are persuadable—those who will respond positively to intervention—versus those who will churn regardless or would have stayed anyway. This enables more efficient resource allocation and prevents wasted effort on ineffective interventions.

Beyond academic interest, this project emphasizes production-readiness. Real-world ML systems require robust data pipelines, comprehensive validation, monitoring for model drift, and explainability for business stakeholders. By building a system that mirrors industry deployment standards, this project bridges the gap between research prototypes and production systems.

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III. Objective / Goals

A. *Technical Goals*

- Achieve AUC $\geq$ 0.85 and F1-score $\geq$ 0.75 for binary churn prediction
- Demonstrate positive uplift scores across customer segments, indicating effective treatment targeting
- Build a modular, scalable data pipeline supporting ingestion, preprocessing, feature engineering, and validation

- Implement meta-learner uplift models (T-learner, S-learner) with comparative evaluation
- Create a validation framework including segmentation analysis, uplift curves, and Qini coefficients

B. *Application Goals*

- Generate actionable, personalized recommendations for at-risk customers
- Provide explainable outputs showing why specific interventions are recommended
- Enable business stakeholders to understand model decisions through feature importance and segment analysis
- Build monitoring capabilities to detect model drift and data quality issues in production scenarios

IV. Scope

A. *Inclusion*

- End-to-end data pipeline (ingestion, preprocessing, feature engineering, validation).
- Binary churn prediction model using gradient boosting (XGBoost/LightGBM)
- Uplift modeling using meta-learner approaches (T-learner, S-learner)
- Recommendation system generating personalized retention actions
- Comprehensive validation framework (segmentation analysis, uplift curves, Qini coefficients)
- Model explainability using SHAP values and feature importance
- Monitoring and drift detection mechanisms
- Documentation covering system architecture, model choices, and deployment considerations.

B. *Exclusions*

- Real-time inference API deployment (system design will be production-ready, but live deployment is out of scope)
- A/B testing framework for live treatment validation
- Integration with external CRM or marketing automation platforms
- Multi-channel recommendation optimization (focus will be on identifying optimal interventions, not channel selection)
- Deep learning approaches (focus on interpretable gradient boosting methods)

V. Proposed Approach / Methodology

A. ***Churn Prediction:*** *Gradient boosting models (XGBoost, LightGBM) will be used for binary classification due to their strong performance on tabular data, ability to handle non-linear relationships, and built-in feature importance. Random Forest will serve as a baseline comparison.*

B. ***Uplift Modeling:*** *Meta-learner approaches will estimate treatment effects:*

- T-learner: Trains separate models for treated and control groups, estimating uplift as the difference in predicted outcomes
- S-learner: Trains a single model with treatment as a feature, estimating uplift by comparing predictions with/without treatment
- Both approaches will be implemented and compared using uplift curves and Qini coefficients.

C. *Data Processing*

Dataset: A customer transaction dataset containing behavioral features (purchase history, engagement metrics, usage patterns), demographic information, and historical churn labels. If real data is unavailable, a synthetic dataset modeled on Kaggle telecom/subscription churn datasets will be generated.

Preprocessing:

- Handle missing values using imputation (mean/median for numeric, mode for categorical)
- Encode categorical variables using one-hot or target encoding
- Normalize/standardize numeric features
- Split data into training (60%), validation (20%), and test (20%) sets.

Feature Engineering:

- Recency, Frequency, Monetary (RFM) features
- Engagement trends (change in usage over time)
- Customer lifetime value estimates
- Time-based features (tenure, days since last interaction)

D. *Evaluation Metrics*

Churn Prediction:

- AUC-ROC: Area under the receiver operating characteristic curve
- F1-score: Harmonic mean of precision and recall
- Precision and Recall: False positive vs. false negative trade-offs

Uplift Modeling:

- Uplift Curve: Cumulative gain from targeting high-uplift customers
- Qini Coefficient: Area between uplift curve and random targeting baseline

- Segmentation Analysis: Uplift performance across customer segments

E. *System Deisgn and Architecture*

a) High Level Architecture - The system follows a modular, pipeline-based architecture with five core components:

1. Data Pipeline Module

- Ingestion: Load raw customer data from CSV/database
- Validation: Schema checks, range validation, missing value detection
- Preprocessing: Cleaning, encoding, normalization
- Feature Engineering: Derived features (RFM, trends, aggregates)
- Output: Clean, feature-rich dataset ready for modeling.

2. Churn Prediction Module

- Input: Preprocessed features from data pipeline
- Training: XGBoost/LightGBM with hyperparameter tuning (grid search/Optuna)
- Validation: Cross-validation, performance metrics (AUC, F1, precision, recall)
- Output: Binary churn predictions + probabilities for each customer.

3. Uplift Modeling Module

- Input: Features + treatment assignment (historical or simulated)
- T-learner: Separate models for treated/control groups
- S-learner: Single model with treatment as feature
- Evaluation: Uplift curves, Qini coefficients, segmentation analysis
- Output: Uplift scores indicating treatment responsiveness.

4. Recommendation Module

- Input: Churn probabilities + uplift scores
- Logic: Prioritize customers with high churn probability AND high uplift
- Intervention Mapping: Match customer profiles to optimal interventions (discount, feature suggestion, outreach)
- Explainability: SHAP values showing key factors driving recommendations
- Output: Ranked list of actionable recommendations per customer.

5. Monitoring & Validation Module

- Data Quality Checks: Schema drift, feature distribution shifts
- Model Performance: Track AUC, F1, uplift over time
- Drift Detection: Statistical tests for input/output drift
- Alerting: Trigger retraining when performance degrades
- Output: Dashboard with system health metrics and alerts.

6. Module Interaction

Data flows sequentially through the pipeline:

Raw Data → Data Pipeline → Churn Prediction + Uplift Modeling → Recommendation Module → Actionable Outputs.

The Monitoring Module runs continuously, observing data quality at ingestion, model performance after prediction, and recommendation distribution. All modules are designed to be independently testable and replaceable, following software engineering best practices for modularity.

VI. Dataset & Resources

A. *Dataset*

Primary Source: Kaggle Telco Customer Churn dataset (7,043 customers, 21 features) or similar subscription-based churn datasets.

Key Features:

- Demographic: Age, gender, location
- Behavioral: Tenure, monthly charges, contract type, payment method
- Engagement: Service usage, support calls, complaint history
- Target: Binary churn label (Yes/No)

Treatment Data: For uplift modeling, treatment assignment will be simulated based on historical patterns (e.g., customers who received discounts vs. those who did not). If available, real A/B test data will be used.

Licensing: Public domain datasets from Kaggle are licensed for educational and research use.

B. *Compute Resources*

Local Machine: Primary development and testing (16GB RAM, multi-core CPU)

HiPerGator (if needed): For large-scale hyperparameter tuning and ensemble training

Cloud Resources (optional): Google Colab or AWS SageMaker for GPU-accelerated experiments.

Given the dataset size (~7K rows), local compute should be sufficient. Cloud resources will be used only if scaling to larger datasets or exploring deep learning approaches.

VII. Implementation and Timeline

Week	Milestone	Deliverables
1-2	Project Setup & Literature Review	Finalized project plan, dataset acquisition, literature review
3	Data Pipeline Development	Data ingestion, validation, preprocessing scripts
4	Feature Engineering	Engineered features (RFM, trends, aggregates)
5	Churn Prediction Model	Trained XGBoost/LightGBM, baseline performance metrics
6	Model Optimization	Hyperparameter tuning, cross-validation, AUC ≥ 0.85
7	Uplift Modeling - T-learner	T-learner implementation, initial uplift scores
8	Uplift Modeling - S-learner	S-learner implementation, comparative evaluation
9	Recommendation Module	Intervention mapping, SHAP explainability
10	Validation & Testing	Segmentation analysis, uplift curves, Qini coefficients
11	Monitoring & Documentation	Drift detection, system documentation, final report draft
12	Final Presentation & Submission	Project demo, final report, presentation

VIII. Evaluation and Test Planning

A. *Validation Strategy*

Train-Validation-Test Split: 60% training, 20% validation, 20% test. Test set remains untouched until final evaluation.

K-Fold Cross-Validation: 5-fold stratified CV on training set to ensure robust performance estimates.

Temporal Validation: If data has timestamps, use time-based splits to simulate production scenarios (train on past, test on future).

Segmentation Testing: Evaluate model performance across customer segments (high-value, new users, long-tenured) to identify biases.

Uplift Validation: Use uplift curves and Qini coefficients to validate that high-uplift customers truly respond better to interventions.

Sanity Checks: Verify that recommendations align with domain knowledge (e.g., don't recommend expensive interventions to low-value customers).

Backtracking: Trace model outputs back to input features to ensure decisions are based on valid patterns, not data leakage.

B. *Performance Metrics*

1. Churn Prediction Metrics:

- AUC-ROC ⩾ 0.85 (discriminative ability)
- F1-Score ⩾ 0.75 (balance between precision and recall)
- Precision (minimize false positives to avoid wasted interventions)
- Recall (minimize false negatives to catch at-risk customers)

2. Uplift Modeling Metrics:

- Qini Coefficient > 0.1 (area between uplift curve and random baseline)
- Uplift@10% (uplift captured by targeting top 10% of customers)
- Segment-Level Uplift: Positive uplift across major customer segments

3. Business Metrics:

- Intervention Efficiency: % of recommendations that align with domain expertise
- Explainability Score: Qualitative assessment of SHAP value interpretability
- System Latency: Inference time per customer (target < 100ms for production scenarios)

IX. Risk and Mitigation Strategies

A. *Potential Challenges*

1. Data Quality Issues

Risk: Missing values, outliers, or imbalanced classes degrade model performance.

Mitigation: Implement robust data validation checks. Use imputation techniques and SMOTE/undersampling for class imbalance. Maintain detailed data quality logs.

2. Limited Treatment Data for Uplift Modeling

Risk: Insufficient historical treatment data makes uplift estimation unreliable.

Mitigation: Simulate treatment assignments based on observational data patterns. Use propensity score

matching to create pseudo-treatment groups. Consider synthetic data augmentation.

3. Model Overfitting

Risk: Complex models memorize training data rather than learning generalizable patterns.

Mitigation: Use regularization (L2 in XGBoost), early stopping, and cross-validation. Monitor validation vs. test performance. Implement feature selection to reduce dimensionality.

4. Computational Resource Limitations

Risk: Large-scale hyperparameter tuning or ensemble methods exceed local compute capacity.

Mitigation: Use HiPerGator for intensive tasks. Leverage cloud platforms (Google Colab, AWS) if needed. Optimize code with parallel processing and efficient data structures.

5. Explainability Challenges

Risk: Stakeholders may not trust black-box uplift models without clear explanations.

Mitigation: Use SHAP values for feature-level explanations. Create visualizations showing top drivers of churn/uplift. Provide segment-level summaries and case studies.

6. Time Constraints

Risk: Ambitious scope may lead to incomplete implementation within the semester.

Mitigation: Prioritize core functionality (churn prediction + basic uplift). Treat advanced features (multi-treatment optimization, deep learning) as stretch goals. Maintain weekly progress tracking.

X. Literature Review and Related Work

A. *Existing Solution*

Churn Prediction Models: Traditional approaches use logistic regression, random forests, or gradient boosting for binary classification [1]. Recent work has explored deep learning with LSTM networks for temporal patterns [2]. However, these models focus solely on prediction accuracy without considering treatment effects.

Uplift Modeling Techniques: Meta-learner approaches (T-learner, S-learner, X-learner) have emerged as practical methods for estimating heterogeneous treatment effects [3][4]. The T-learner trains separate models for treatment and control groups, while the S-learner uses a single model with treatment as a feature. Empirical studies show T-learner performs well with sufficient data, while S-learner is more data-efficient but less flexible [5].

Production ML Systems: Industry best practices emphasize data validation, model monitoring, and drift detection [6]. Google's TFX and similar frameworks provide reference architectures for production ML pipelines [7]. Key principles include versioned data, reproducible training, automated validation, and continuous monitoring.

B. *Gap Analysis*

While churn prediction and uplift modeling have been studied independently, few projects integrate them into a unified, production-ready system. Existing research often treats uplift modeling as a standalone technique without addressing practical deployment challenges such as data pipeline design, validation beyond metrics, and business-facing explainability.

C. *This project addresses these gaps by:*

- Integrating churn prediction and uplift modeling into a single end-to-end system
- Emphasizing production-style architecture with robust data pipelines and validation
- Providing actionable recommendations with business-oriented explainability
- Implementing validation mechanisms that go beyond standard metrics to assess real-world effectiveness

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